5 November 2026

Editorial Office
Measurement

Dear Editor,

Please consider our manuscript, "Pooled Accuracy Is Not Enough to Authorize Fixed-K3 VMD Preprocessing: Same-Source and External Bearing Audits," for publication in *Measurement*.

The manuscript addresses a measurement-evaluation problem: a preprocessing decision is often authorized by pooled diagnostic accuracy even when measurement conditions, validation units, and class-specific effects can change what that number means. The paper's contribution is an empirical audit of this authorization problem, not a new VMD algorithm or classifier.

The first central result is a same-source HUSTbearing audit. Against a clean-trained raw baseline, fixed K3 shows a speed-level gain of 9.72 percentage points. Under the evaluated fixed matched 0 dB noise realization, the gain reverses sign, yielding a matched effect of -7.94 points (unit-only interval [-12.61, -3.44]; crossed robustness interval [-12.72, -3.28]). The matched-augmentation audit is Not qualified under an illustrative materiality margin of 1 point. The clean-trained advantage is therefore conditional on the evaluated noise configuration, not a deployment warrant.

The second central result is an independent PHM2012/PRONOSTIA run-to-failure audit with 17 non-fragile physical bearings. Clean-trained K3 raises pooled balanced accuracy by 24.51 points, but the healthy class improves by 70.75 points while the degraded class falls by 21.73 points (simultaneous interval [-43.14, -5.39]). The block is therefore Not qualified on class-harm grounds despite its positive pooled mean. Matched augmentation changes the mean effect to -6.29 points. This external block makes the central measurement point visible: a pooled improvement can coexist with class-specific harm.

The study is a secondary analysis of completed vibration experiments and adds no physical test campaign. Additional HSE, HUST-Hanoi, SCA, and Mehran-UET audits remain Insufficient or Not qualified under the same evidentiary discipline. A three-state evidence audit is used to distinguish Not qualified from Insufficient and to prevent a missing or fragile class interval from being read as a pass. This audit is a decision aid, not a validated certification framework, and no real block receives Qualified in the present evidence. It also reports the evidence gap behind each Insufficient result, separating established failure from missing or fragile evidence that additional independent units could in principle resolve. CCMG is reported only as an optional measurement-condition prefilter with small incremental value.

The work directly addresses two areas in the journal's scope: evaluation procedures for performance analysis of measurement processes and algorithms, and measurement data processing. The cover letter does not position the paper as a new sensor-calibration method or a new fault-diagnosis network.

The manuscript is original, is not under consideration elsewhere, and all authors will approve the submitted version. The analysis code, frozen protocols, generated tables, figures, and result files are available at https://github.com/KKK-cell441/fixed-k3-pooled-accuracy-audit. The source datasets retain their original licenses and access conditions.

Sincerely,
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